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| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Merge Sort
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Combining sorted sublists
- c. Dividing the input into smaller subproblems
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Merge Sort
- c. Binary Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 200,470,80,150,40,90,300,400,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

**The process of placing or rearranging a collection of elements into a particular order is known as**

a. Searching
b. Sorting
c. Merging
d. Rearranging

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It has no effect on the search operation
c. It makes the search operation slower
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **9**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Dividing the list into halves
c. Bubbling up the largest element to its correct position
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It divides the list into two halves, sorts each half, and then merges them
c. It compares adjacent elements and swaps them if necessary
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By comparing elements sequentially
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are split again into smaller sublists
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It requires external libraries
c. It only works with integer arrays
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. sort()
d. order()

The correct answer is: sort()

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Binary Search
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort